

SAT Lookup Sheet: Scalar-Angular-Torsion (SAT) Unified Construct

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Abstract

We present the Scalar-Angular-Torsion (SAT) unified construct, integrating gravity, gauge fields, matter, filament and time-flow dynamics, torsion, and holonomy into a single mathematical framework. The construct encodes particle spectra, cosmology, electroweak interactions, exotic states, materials physics, optics, and extreme classical phenomena.

1 Unified Action

The covariant SAT action is

$$\begin{aligned}
 S_{\text{SAT}}[g_{\mu\nu}, A_\mu, \psi, \theta, u_\mu, J_{\mu\nu}, \tau] = \int d^4x \sqrt{-g} & \left[\underbrace{\frac{1}{2\kappa} R[g]}_{\text{GR curvature}} - \underbrace{\frac{1}{4g_A^2} \text{Tr}(F_{\mu\nu} F^{\mu\nu})}_{\text{Gauge sector}} \right. \\
 & \left. - \underbrace{\frac{1}{2} (\nabla_\mu \theta)(\nabla^\mu \theta) - V(\theta)}_{\text{Mass / coupling field}} - \underbrace{L_{\text{matter}}(\psi, D_A \psi, \theta)}_{\text{Fermion/boson content}} - \underbrace{L_u(u_\mu, \delta u_\mu) + L_J(J_{\mu\nu}, \tau)}_{\text{Filament / time-flow dynamics}} \right] + S_{\text{hol}}[\theta, \tau]
 \end{aligned} \tag{1}$$

Definitions:

- $g_{\mu\nu}$: spacetime metric
- $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu + [A_\mu, A_\nu]$: gauge curvature
- $\theta(x)$: scalar angle encoding mass and coupling via worldline deviation
- ψ : matter fields (fermions, scalars)
- u_μ : unit time-flow vector field
- $J_{\mu\nu}$: filament stress/strain tensor
- τ : discrete torsion/topological index
- S_{hol} : holonomy / winding constraints ($\oint d\phi = 2\pi n$)

2 Particle Worldline and Mass

$$\mathbf{r}_i(t) = \mathbf{r}_{0,i}(t) + \sum_j f_{ij}(t) \hat{e}_j, \quad f_{ij}(t) \propto \sin\left(\frac{n\pi t}{T} + \phi\right) \quad (2)$$

$$m_{\text{eff}} = m_{\text{topo}} + m_{\text{ind}}, \quad m_{\text{topo}} = \frac{m_0}{Q}, \quad m_0 = \frac{T\ell_f}{c^2}, \quad m_{\text{ind}} \approx \frac{Cg_1^2 M^2}{E_{\text{overlap}}} \quad (3)$$

- Topological charge: $Q = L_{\text{wind}} + L_{\text{link}} + W_{\text{writhe}}$
- Quadratic excited states: $m^2(N, Q) = k(N + \beta Q^2 - a) e^{\gamma_\chi \tau_\chi}$
- Electron, neutrino, and hadron masses calculated with appropriate Q and torsion τ_χ

3 Quantum and Holonomy Structure

$$[\hat{\xi}_\mu(\lambda), \hat{\pi}_\nu(\lambda')] = i\hbar \delta_{\mu\nu} \delta(\lambda - \lambda'), \quad (4)$$

$$[\phi_a(x), \pi_b(y)] = i\hbar \delta_{ab} \delta^3(x - y), \quad (5)$$

$$\oint_{\mathcal{C}} d\phi = 2\pi n, \quad n \in \mathbb{Z}, \quad n \leq 3 \text{ for stability} \quad (6)$$

$$J = \cos \theta_{12} \sin \theta_{12} \cos \theta_{23} \sin \theta_{23} \cos^2 \theta_{13} \sin \theta_{13} \sin \delta_{CP}, \quad (7)$$

$$\delta_{CP} = n_A \Phi_A + n_B \Phi_B \pmod{\pi}, \quad (8)$$

$$m_\nu = \{5.0, 8.9, 16.3\} \text{ meV}, \quad \sum m_\nu = 0.03 \text{ eV}, \quad (9)$$

$$|m_{\beta\beta}| \in \{0.40, 1.12, 5.69, 6.41\} \text{ meV} \quad (10)$$

4 Filament and Time-Flow Dynamics

$$\mathcal{R}(r) = \frac{\rho_{\text{link}}(r)}{|\cos \theta_4(r)|} + \epsilon(1 + \alpha\tau(r)), \quad (11)$$

$$H(t) = \frac{1}{R(t)} \frac{dR}{dt} = \frac{1}{R(t)\mathcal{R}(R(t))}, \quad (12)$$

$$\rho(\partial_t v + (v \cdot \nabla)v) = -\nabla P - \rho \nabla W(\Phi) - \kappa(\nabla^2 \Phi) \nabla \Phi + \lambda \nabla(\rho \Phi), \quad (13)$$

$$\Gamma = \Lambda \exp \left[-\frac{\alpha_{\text{top}}^2}{E_{\text{conf}}} \right], \quad E_{\text{conf}} = m(Q) - \sum_i m(Q_i) \quad (14)$$

5 Cosmology and Gravity

$$r = \frac{\int (\delta\rho_{\text{link}})^2 d\Omega_{\text{lateral}}}{\int (\delta\theta_4)^2 d\Omega_{\text{radial}}} \approx 0.011, \quad (15)$$

$$\Delta\psi = g_h \int_{\text{emit}}^{\text{obs}} \Theta(s) ds \approx g_h \Theta_0 L_{\text{eff}}, \quad g_h \Theta_0 \lesssim 3.74 \times 10^{-7}, \quad (16)$$

$$G_{\text{ind}} \sim \frac{c^4 \ell_f^2}{16\pi A \kappa}, \quad G_{\text{eff}} \sim G_{\text{ind}} \text{ in low-energy limit}, \quad (17)$$

$$S_{\text{eff}} = \frac{T}{16\pi} \int d^4x \sqrt{-g} (R + \beta R^2), \quad \beta = \frac{2}{6\pi} \quad (18)$$

6 High-Energy and Exotic States

$$m_B = 0.74 \text{ TeV}, \quad \Gamma_B < 10^{-6} \text{ MeV}, \quad \Omega h^2 = 0.012, \quad (19)$$

$$\sigma_{\text{MBH}}(100 \text{ TeV}) \approx 0.077 \text{ fb}, \quad \sigma_{\text{MBH}}(14 \text{ TeV}) < 3.6 \times 10^{-3} \text{ fb}, \quad (20)$$

$$R = 0.007304 \approx \alpha \quad (21)$$

7 Materials and Optical Effects

$$\theta_4(x) = \frac{2\pi}{3} \frac{1 + \tanh(\mu x)}{2}, \quad \Delta\phi = \int \Delta n(x) k dx, \quad \Delta n(x) = \eta \sin^2(\theta_4(x)), \quad (22)$$

$$\tau_1 + \tau_2 + \tau_3 \equiv 0 \pmod{3}, \quad (23)$$

$$\Theta_D = \frac{\hbar}{k_B} v_s (6\pi^2 n)^{1/3}, \quad T_m = \frac{c_L^2 M a^2 k_B}{9\hbar^2} \Theta_D^2 \quad (24)$$

8 Extreme and Everyday Physics

$$\Delta A = \frac{8\pi}{\kappa} \mathcal{E}_1, \quad S = \frac{A}{4} = n, \quad (25)$$

$$\sigma_\theta = \frac{pr}{t}, \quad \sigma_y t = \frac{\rho g R r}{2}, \quad (26)$$

$$R = \frac{v_0^2}{g} \sin 2\theta, \quad (27)$$

$$\Gamma_{n \rightarrow n-1} \propto e^{-\Delta S} = e^{-\Delta A/4}, \quad \Delta f = \frac{\Delta E}{h} = \frac{c}{8\pi M} \quad (28)$$

9 Summary

1. The SAT action encodes GR, gauge fields, matter, filaments, torsion, and holonomy.
2. Mass spectra emerge from topological charge Q , torsion τ_χ , and induced inertial corrections.
3. Quantum mechanics is emergent from holonomy of particle worldlines.

4. Cosmology and gravity arise from filament intersection density and projection angles.
5. Exotic states and high-energy phenomena are topologically constrained.
6. Condensed matter and optics follow from filament/torsion rules.
7. Everyday physics (mechanics, black holes, planetary atmospheres) is derivable from the same action.